

K-FIELD

CELL3 TO SPECTROPHOTOMETER PLANAR COMPARISON

Authoritative Cell3 orientation, reciprocal planes, recursive hierarchy, high-index zone structure, measured harmonic branches, and the exposed direct-space base face

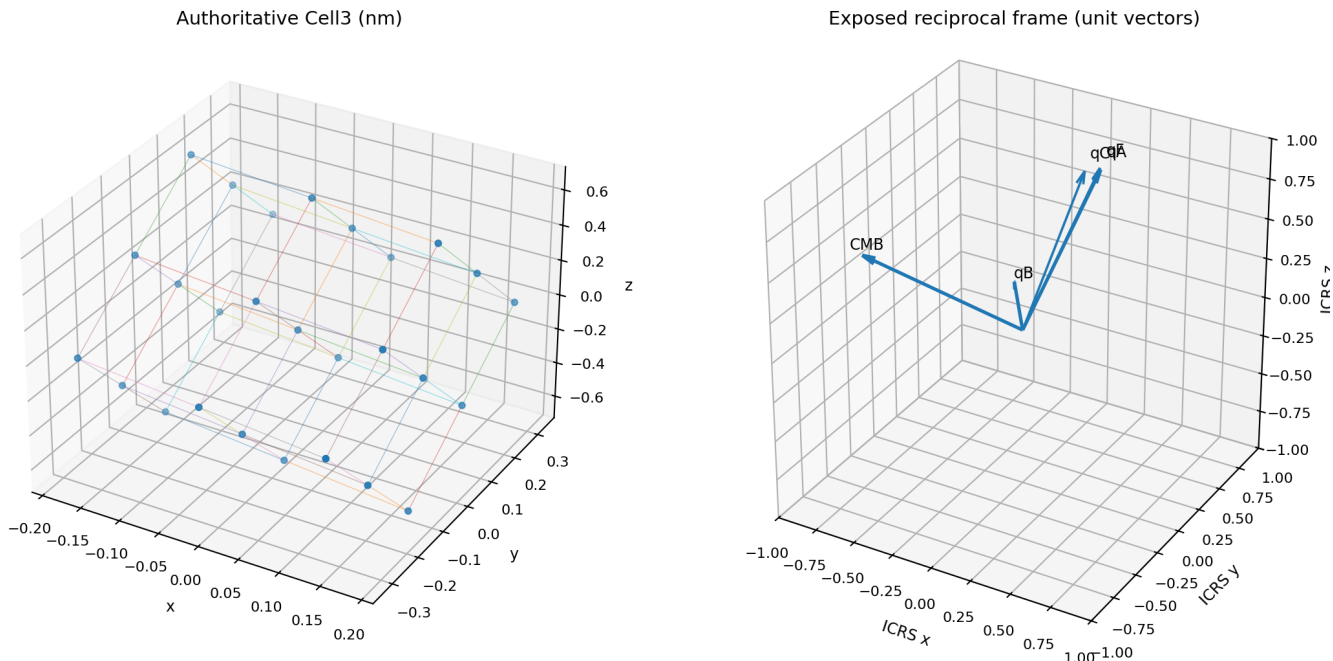


Figure 1. Data-derived title composite. Left: the authoritative 27-node Cell3 primitive volume in ICRS nanometres. Right: the pic_id14E reciprocal plane axes qA, qB, qC, qF and the supplied CMB direction in the same ICRS orientation.

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Abstract

This archival reference compares the precisely recovered pic_id14E spectrophotometer plane structure with the authoritative Cell3 direct and reciprocal geometry. The comparison is performed in one absolute International Celestial Reference System frame. It distinguishes the primitive Cell3 basis, the reciprocal plane atlas, the recursive Cell3 hierarchy, measured plane normals, harmonic order, direct-space duals, and integer zone relationships. The principal result is that the measured planes are not the primitive Cell3 coordinate faces. They form a high-index, recursively enlarged Cell3 zone whose intersection follows a CMB-like longitudinal direction.

The measured spectral-polarization plane qA has spacing 317.864281 km at RA 81.380625 deg and Dec +45.795586 deg. The angular-variance plane qB has spacing 1,034.800487 km at RA 296.484362 deg and Dec +44.633516 deg. Their axial separation is 84.391343 deg, and their cross product is 0.036386 deg from the supplied CMB axis. A bridge carrier qF lies 2.795911 deg from qA and has spacing 670.885616 km. The ratio dF/dA is 2.110603975, differing from 19/9 by only -0.02402 percent. Both qA and qF are reproduced by the high-index Cell3 family (1,6,19) at recursive level r=18 with harmonic orders 19 and 9.

The qA and qB intersection has Cell3 direct-coordinate ratio [1, 0.28506983, -0.13159608], almost identical to the supplied CMB ratio [1, 0.28464349, -0.13158840]. The lowest-order nearby integer direction is [7,2,-1], which yields the zone law $7h + 2k - l = 0$. The exact zone assignments (1,6,19) and (2,-9,-4) both satisfy this law. The independently extracted broadband mode qC obeys qC approximately qA + 2qB. A combination-optimized Cell3 construction produces the integer reciprocal packet (17,-258,-433), predicting qC direction to 0.659366 deg and spacing to 0.116218 percent.

The measured reciprocal pair also defines a dimensionally recovered transverse direct-space base face. Its edge lengths are 319.393331 km and 1,039.778275 km, its included angle is 95.608657 deg, and its area is $3.305083742 \times 10^{11} \text{ m}^2$. The harmonic Cell3 construction predicts these quantities to approximately 0.1-0.8 percent in length and ratio, 0.500 percent in area, and 0.003723 deg in included angle. The longitudinal period along the CMB-like zone axis remains unresolved. The exposed structure therefore supplies a measured two-dimensional Cell3 reciprocal zone and direct-space face, but not a closed three-dimensional primitive cell.

Central result. pic_id14E exposes a high-index, recursively enlarged Cell3 zone, not a direct macroscopic copy of the primitive Cell3 coordinate faces. The observed plane spacings are organized by the Cell3 9^{th} hierarchy, low-integer harmonic order, a common integer zone equation, and reciprocal-vector sum closure.

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1. Purpose, Scope, and Geometric Language

The purpose of this reference is to compare two independently specified geometric objects in one coordinate frame. The first is the authoritative Cell3 basis, fixed in ICRS coordinates and measured in nanometres. The second is the large-scale plane structure recovered from the full-valid pic_id14E rotating spectrophotometer record. The central questions are orientation, plane spacing, ratios, harmonic order, integer geometry, and direct-space reconstruction.

A direct lattice basis describes translations between nodes. A reciprocal vector describes a family of parallel planes. The two objects have inverse dimensions. A direct vector has units of length; a reciprocal vector has units of cycles per length. The plane spacing is the reciprocal-vector magnitude inverted.

```
Direct position:  $x = B u$ 
Reciprocal plane vector:  $g_{hkl} = B^{(-T)} [h \ k \ l]^T$ 
Plane equation:  $g_{hkl} \cdot x = \text{integer}$ 
Plane spacing:  $d_{hkl} = 1 / ||g_{hkl}||$ 
```

A plane family is axial: reversing its normal leaves the same physical planes. Angular comparisons therefore use the smaller angle between either normal direction. A zone is a collection of planes sharing one direct-space line. If the direct zone direction is $[u,v,w]$, a plane (h,k,l) belongs to that zone when $hu + kv + lw = 0$.

1.1 Observed, derived, and inferred quantities

Observed/recovered: qA , qB , qC , qF , their ICRS orientations, measured spacings, and the dimensionally recovered dual face.

Derived: reciprocal Cell3 coordinates, zone direction, recursive harmonic predictions, direct-space dual vectors, and vector-sum packets. **Geometric interpretation:** a high-index recursive Cell3 zone with multiple harmonic branches. **Unresolved:** the longitudinal period and absolute three-dimensional registry along the zone axis.

2. Authoritative Cell3 Direct Geometry

The authoritative ordered Cell3 basis B is fixed directly in ICRS Cartesian coordinates. Its columns are the direct vectors a, b, and c. No fitted global rotation, axis permutation, or declination inversion is applied in this comparison.

```
B_nm = [[ 0.327501242056, 0.035330775844, -0.010129317627 ],
[ -0.054897007884, 0.207319850440, -0.389787702366 ],
[ -0.037487502895, -0.547401085389, -0.689734408367 ]] nm
```

Vector	ICRS x (nm)	ICRS y (nm)	ICRS z (nm)	Length (nm)	RA (deg)	Dec (deg)
a	0.327501242056	-0.054897007884	-0.037487502895	0.334179679059	350.484328	-6.440865
b	0.035330775844	0.207319850440	-0.547401085389	0.586410890412	80.328749	-68.983472
c	-0.010129317627	-0.389787702366	-0.689734408367	0.792319765042	268.511404	-60.519741

The ordered basis is left-handed because $\det(B) = -0.117914891912 \text{ nm}^3$. Its absolute primitive volume is $0.117914891912 \text{ nm}^3$. The edge lengths are 0.334179679059, 0.586410890412, and 0.792319765042 nm.

2.1 Primitive metric

Metric	Value
angle(a,b)	83.933483012249 deg
angle(a,c)	80.448130018322 deg
angle(b,c)	50.363231813718 deg
A_ab	0.194869165461 nm ²
A_ac	0.261106235790 nm ²
A_bc	0.357809539128 nm ²

Authoritative Cell3: all 27 centered half-step nodes

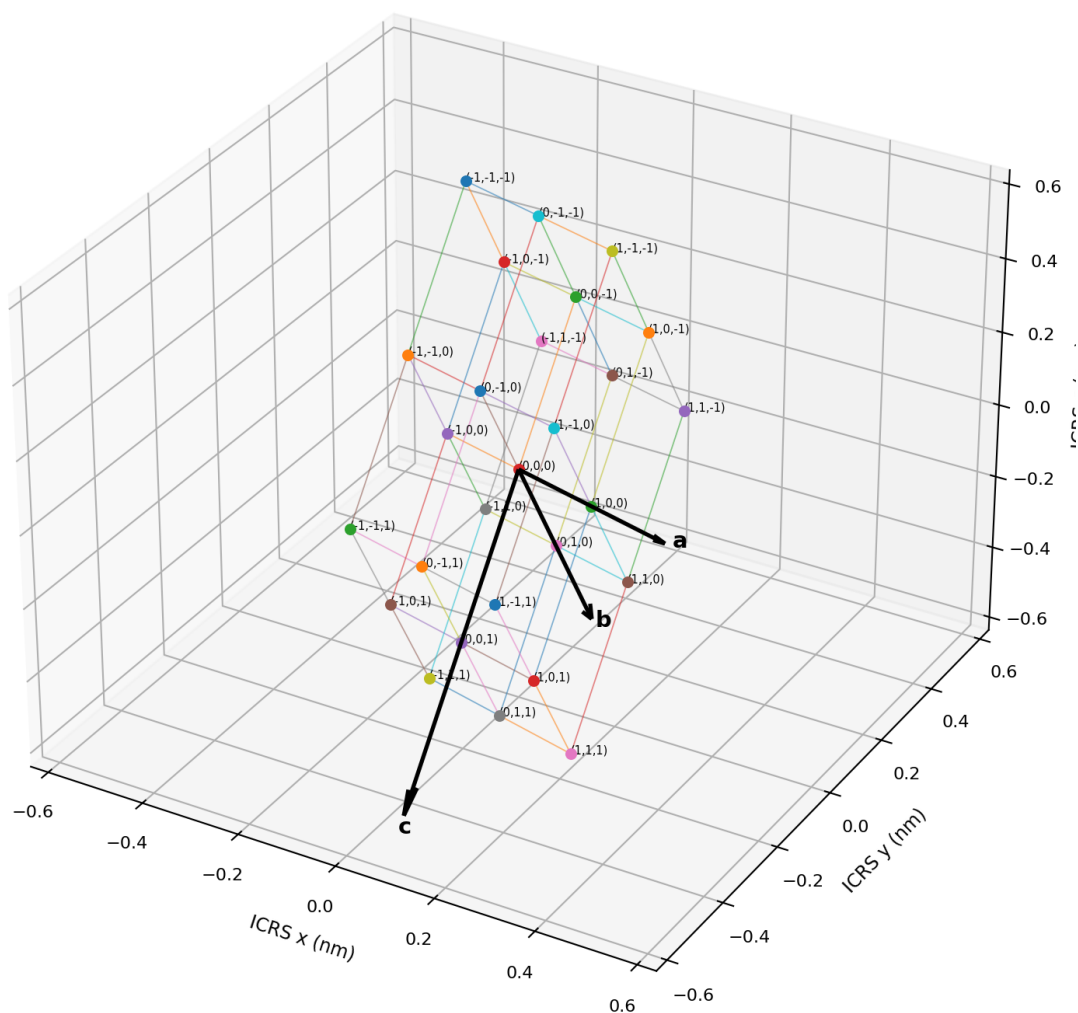


Figure 2. Authoritative Cell3 with all 27 centered half-step nodes. Node addresses are (i,j,k), where the physical fractional coordinate is (i,j,k)/2. Nearest-neighbor links differ by one address unit in exactly one coordinate. The basis arrows a, b, and c are plotted at their exact nanometre lengths and ICRS orientations.

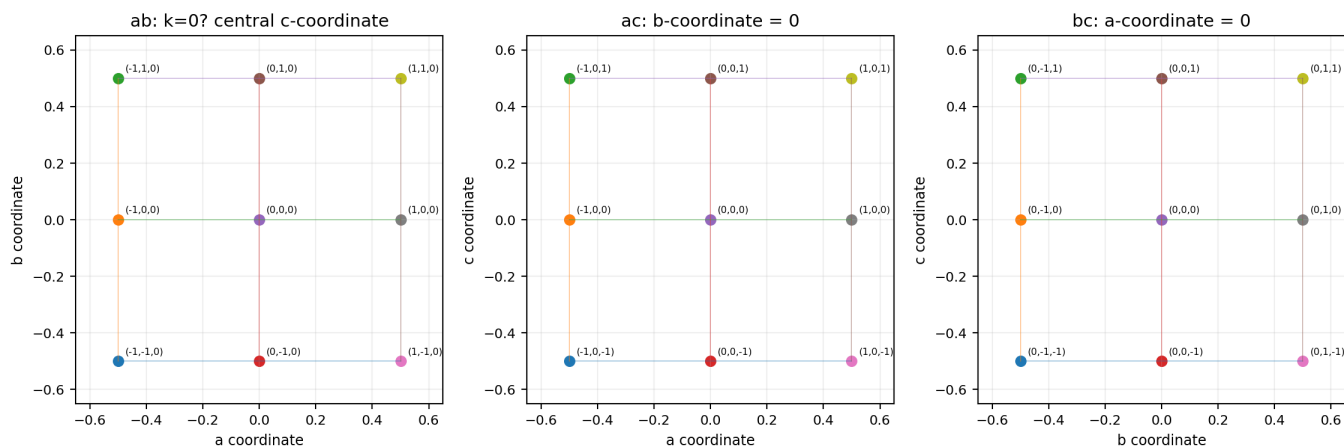


Figure 3. All nine nodes on each central coordinate plane of the authoritative 27-node Cell3: ab, ac, and bc. The axes are fractional direct coordinates. Every plotted node is generated from the authoritative basis and an address in $\{-1,0,+1\}^3$.

The 27-node representation is not an additional fitted geometry. It is the centered half-step sampling of one authoritative primitive volume. It supplies the exact node and face framework from which all indexed planes, recursive copies, and zone directions are computed.

3. Authoritative Cell3 Reciprocal Geometry

The reciprocal basis is $G = B^{\wedge}(-T)$. Its columns are the primitive plane-normal vectors in cycles per nanometre. For an integer packet (h,k,l) , the physical plane normal is $G(h,k,l)^{\wedge}T$, and the adjacent-plane spacing is the inverse magnitude.

$$\begin{aligned} G_{nm}^{-1} &= B^{\wedge}(-T) \\ g_{hkl} &= G [h \ k \ l]^{\wedge}T \\ d_{hkl_nm} &= 1 / ||g_{hkl}|| \end{aligned}$$

Plane	Spacing (nm)	RA (deg)	Dec (deg)	Unit normal x	Unit normal y	Unit normal z
(100)	0.329546529696	355.201792	+1.869279	0.995965190	-0.083602163	0.032619291
(010)	0.451597379721	84.132672	-29.408563	0.089052594	0.866576769	-0.491033948
(001)	0.605097741518	258.000576	-21.000693	-0.194092201	-0.913177169	-0.358379235
(110)	0.266175054578	27.349970	-15.252461	0.856930360	0.443241629	-0.263072646
(101)	0.306442673082	326.873755	-8.694352	0.827844954	-0.540205857	-0.151163370
(011)	0.576944779286	106.780627	-75.703940	-0.071291375	0.236417305	-0.969032712
(111)	0.302545376075	3.082265	-28.568338	0.876976866	0.047223124	-0.478206600

Authoritative low-index Cell3 plane normals versus measured plane axes

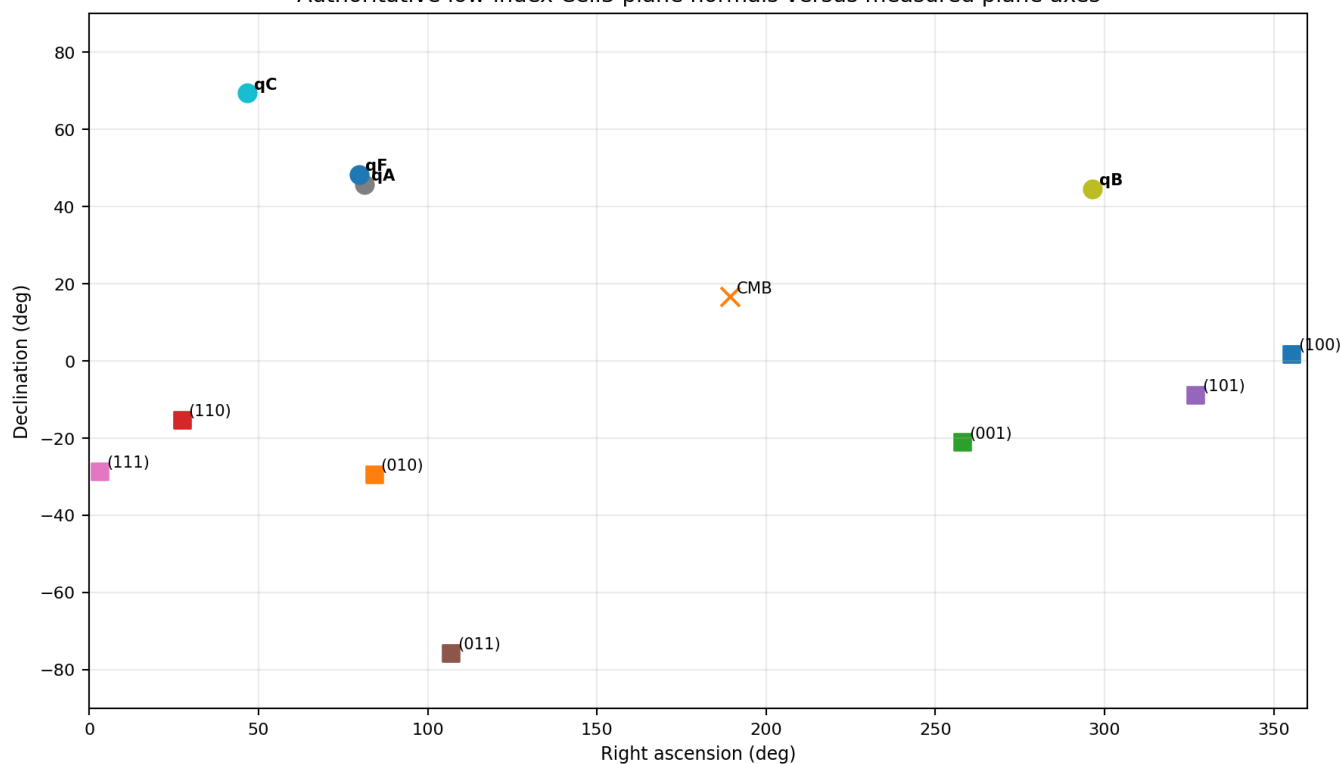


Figure 4. Absolute ICRS orientation of the seven primary Cell3 reciprocal plane families and the measured qA, qB, qC, and qF axes. The large angular separations show that the measured axes are not direct macroscopic copies of the primitive low-index planes.

3.1 Cell3 phase closure

Cell3 diagonal planes are exact reciprocal-vector sums. For example, $g_{110} = g_{100} + g_{010}$, $g_{101} = g_{100} + g_{001}$, and $g_{111} = g_{100} + g_{010} + g_{001}$. This addition law is the reciprocal-space foundation for interpreting the measured qC approximately qA + 2qB relation.

4. Precisely Recovered Spectrophotometer Plane Frame

The pic_id14E full-valid analysis resolves four useful reciprocal modes. qA is associated with the clean spectral-polarization PC2 mode. qB is associated with the 31-position angular-variance PC1 mode. qC is associated with the clean broadband PC1 mode. qF is the strongest continuous 445-Clear-NIR bridge carrier. All are reported in absolute ICRS cycles per metre.

Mode	Spacing (km)	RA (deg)	Dec (deg)	Angle to CMB axis (deg)	Primary observable
qA	317.864281	81.380625	+45.795586	89.984183	spectral PC2
qB	1034.800487	296.484362	+44.633516	89.965843	31-position angular-variance PC1
qC	260.276750	46.645302	+69.405455	89.996828	broadband PC1
qF	670.885616	79.653162	+48.332024	89.983619	445/Clear/NIR bridge carrier

qA and qB are separated by 84.391342976938 deg. Their cross product is 0.036385968967 deg from the supplied CMB axis. qF is 2.795910671289 deg from qA. These relationships define a nearly CMB-transverse two-plane frame with one near-common qA/qF direction.

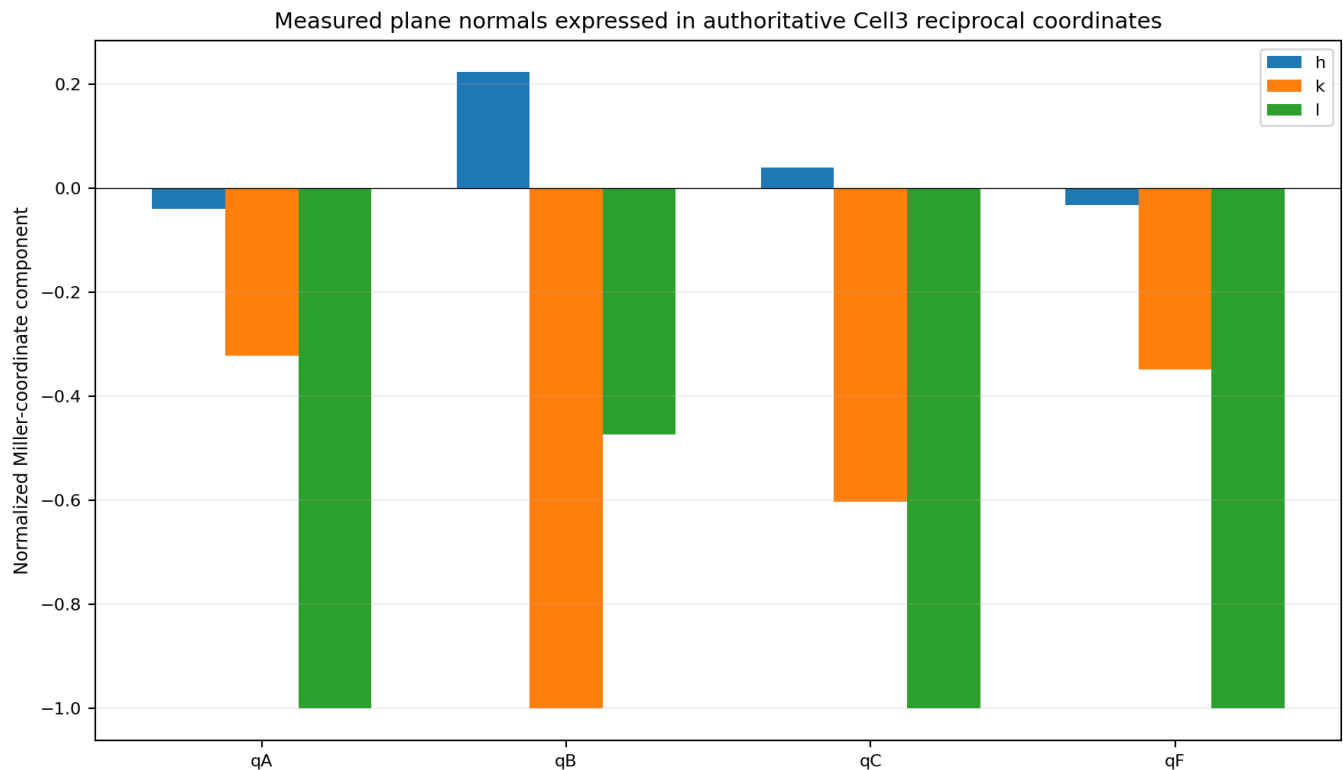


Figure 5. Measured plane normals expressed in authoritative Cell3 reciprocal coordinates. Each triplet is $B^T n$, normalized so the largest absolute component equals one. The nontrivial component ratios demonstrate that the measured planes are high-index oblique families rather than primitive coordinate planes.

4.1 Normalized Cell3 Miller directions

Mode	Normalized h	Normalized k	Normalized l
qA	-0.039904420	-0.321646303	-1.000000000
qB	+0.222703711	-1.000000000	-0.473920797
qC	+0.040055461	-0.602920299	-1.000000000
qF	-0.032152027	-0.348869901	-1.000000000

5. Explicit Comparison with Primary Cell3 Planes

A measured plane is first compared to the seven primary Cell3 families without allowing recursive scaling or high Miller indices. Recursive scaling changes spacing but not direction, so a large angular mismatch cannot be repaired by choosing a different recursive level.

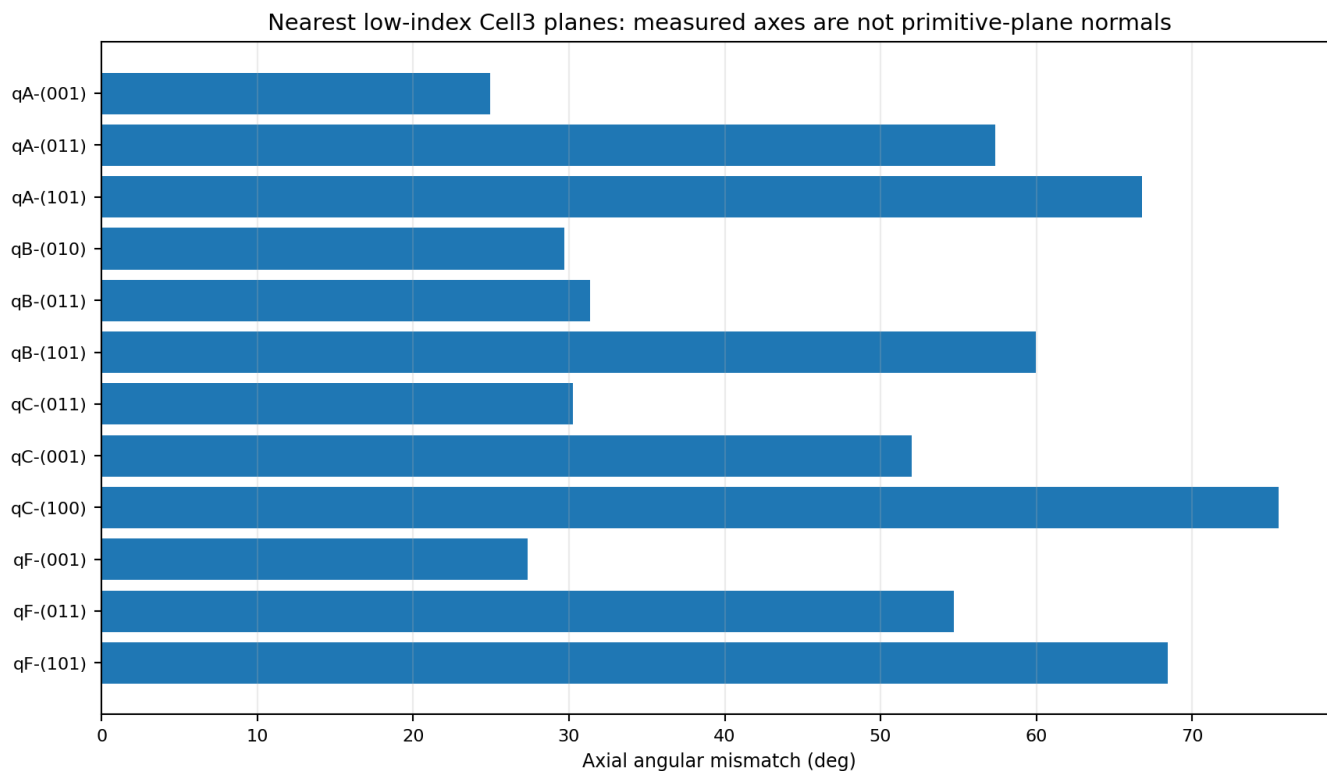


Figure 6. Three nearest low-index Cell3 plane families for each measured axis. Even the closest angular matches are approximately 25-30 deg for qA, qB, qC, and qF. The measured planes are therefore not primitive low-index plane normals.

Measured mode	Closest primary Cell3 plane	Axial mismatch (deg)
qA	(001)	24.949143867
qB	(010)	29.694580214
qC	(011)	30.248009451
qF	(001)	27.363528175

The closest primary matches are qA to (001) at 24.949144 deg, qB to (010) at 29.694580 deg, qC to (011) at 30.248009 deg, and qF to (001) at 27.363528 deg. The direct conclusion is negative but structurally useful: the exposed planes must be sought in the high-index Cell3 reciprocal atlas.

Interpretive rule. A spacing harmonic cannot substitute for an orientation match. The comparison requires both a reciprocal direction and a scale law.

6. Common Cell3 Zone and the CMB-Like Longitudinal Axis

Two nonparallel reciprocal planes intersect along one direct-space line. Transforming the measured $qA \times qB$ line into Cell3 direct coordinates gives a simple ratio that is nearly identical to the supplied CMB direction transformed through the same authoritative basis.

```
Measured intersection in Cell3 coordinates = [1, 0.28506983, -0.13159608]
CMB direction in Cell3 coordinates = [1, 0.28464349, -0.13158840]
Nearest low-order integer direction = [7, 2, -1]
```

The physical Cell3 direction $7a + 2b - c$ lies 1.356368 deg from the supplied CMB axis and 1.377533 deg from the measured $qA \times qB$ intersection. The measured intersection itself is 0.036386 deg from CMB.

Common Cell3 zone: measured plane intersection and CMB-like axis

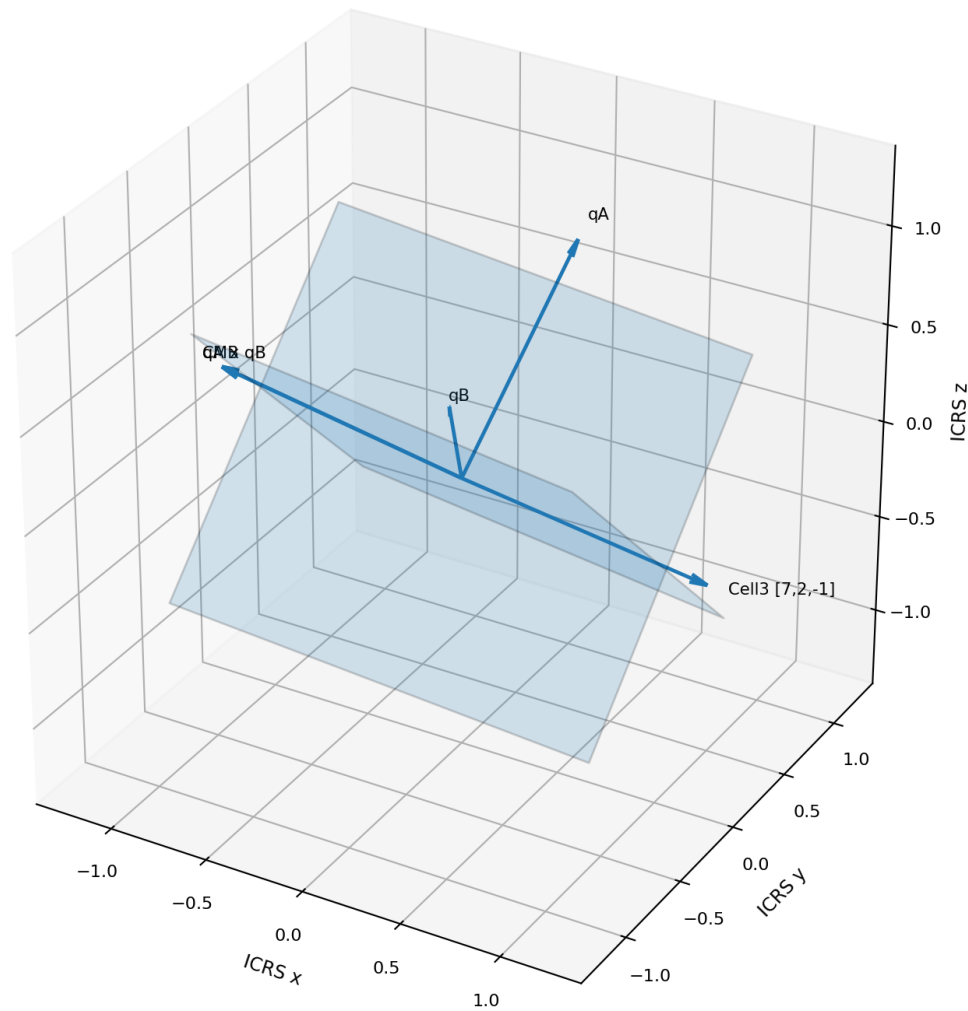


Figure 7. Common Cell3 zone geometry. The qA and qB plane patches intersect along $qA \times qB$. The authoritative direct-space direction $[7,2,-1]$ and the supplied CMB axis nearly coincide with that measured intersection.

6.1 Zone equation

A reciprocal plane (h,k,l) contains the direct direction $[7,2,-1]$ when its dot product with that direction vanishes.

$$7h + 2k - l = 0$$

The high-index family $(1,6,19)$ satisfies $7 + 12 - 19 = 0$. The qB zone-preserving family $(2,-9,-4)$ satisfies $14 - 18 + 4 = 0$. The two independently measured transverse axes can therefore be assigned to one exact authoritative Cell3 zone.

7. Recursive Cell3 Scaling and Harmonic Plane Order

The authoritative Cell3 hierarchy scales the direct basis by powers of nine: $B_r = 9^r B$. A reciprocal vector therefore scales as $9^{(-r)}$, and a plane spacing scales as 9^r . If a measured response occupies harmonic order m of an indexed plane family, its spacing is divided by m .

$$\begin{aligned} B_r &= 9^r B \\ G_r &= 9^{(-r)} G \\ q_{hkl}(r,m) &= (m / 9^r) G [h \ k \ l]^T \times 10^9 \text{ cycles/m} \\ d_{hkl}(r,m) &= 9^r / (m \times 10^9 \ ||G [h \ k \ l]^T||) \text{ m} \end{aligned}$$

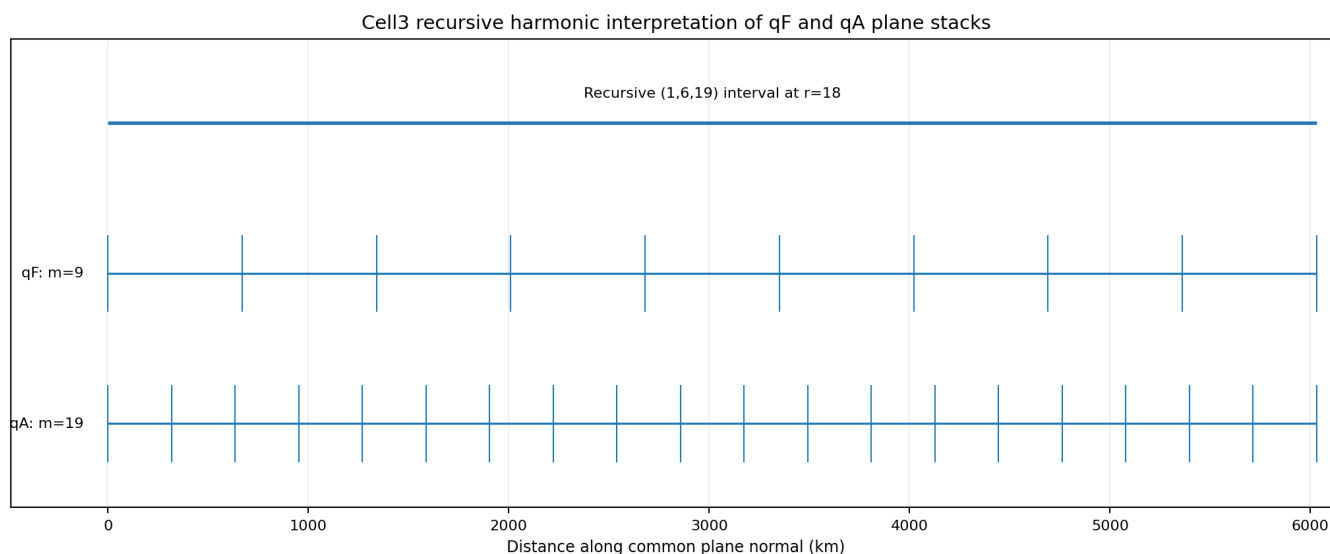


Figure 8. Dimensionally correct sheet-stack representation for the common high-index family (1,6,19) at recursive level $r=18$. The complete recursive interval is 6,031.232391 km. qF divides it into 9 equal plane intervals; qA divides it into 19 equal intervals.

7.1 qA and qF: one high-index family

The primitive Cell3 spacing of (1,6,19) is 0.040182864491 nm. At $r=18$ it becomes 6031.232391 km before harmonic subdivision. Harmonic order $m=19$ predicts qA spacing 317.433284 km, compared with 317.864281 km. Harmonic order $m=9$ predicts qF spacing 670.136932 km, compared with 670.885616 km.

$$\begin{aligned} dF / dA \text{ observed} &= 2.110603974823 \\ 19 / 9 \text{ target} &= 2.111111111111 \\ \text{relative residual} &= -0.024022 \text{ percent} \end{aligned}$$

The $m=9, r=18$ description of qF is algebraically identical to $m=1, r=17$ because one recursive level contributes an exact factor of nine. qF can therefore be read as the fundamental of recursive level 17, while qA is the 19th harmonic of the next 9x direct-space level.

7.2 qB assignments

The exact zone-preserving qB assignment is (2,-9,-4), $r=18, m=8$. It predicts 1,046.724798 km, an orientation residual of 1.360906 deg, and a spacing residual of -1.139202 percent. A second assignment, (1,-4,-2), $r=18, m=18$, predicts 1,041.406943 km with 2.746287 deg orientation residual and -0.634378 percent spacing residual. The second assignment is used for the tight qC combination packet; the first preserves the exact [7,2,-1] zone equation.

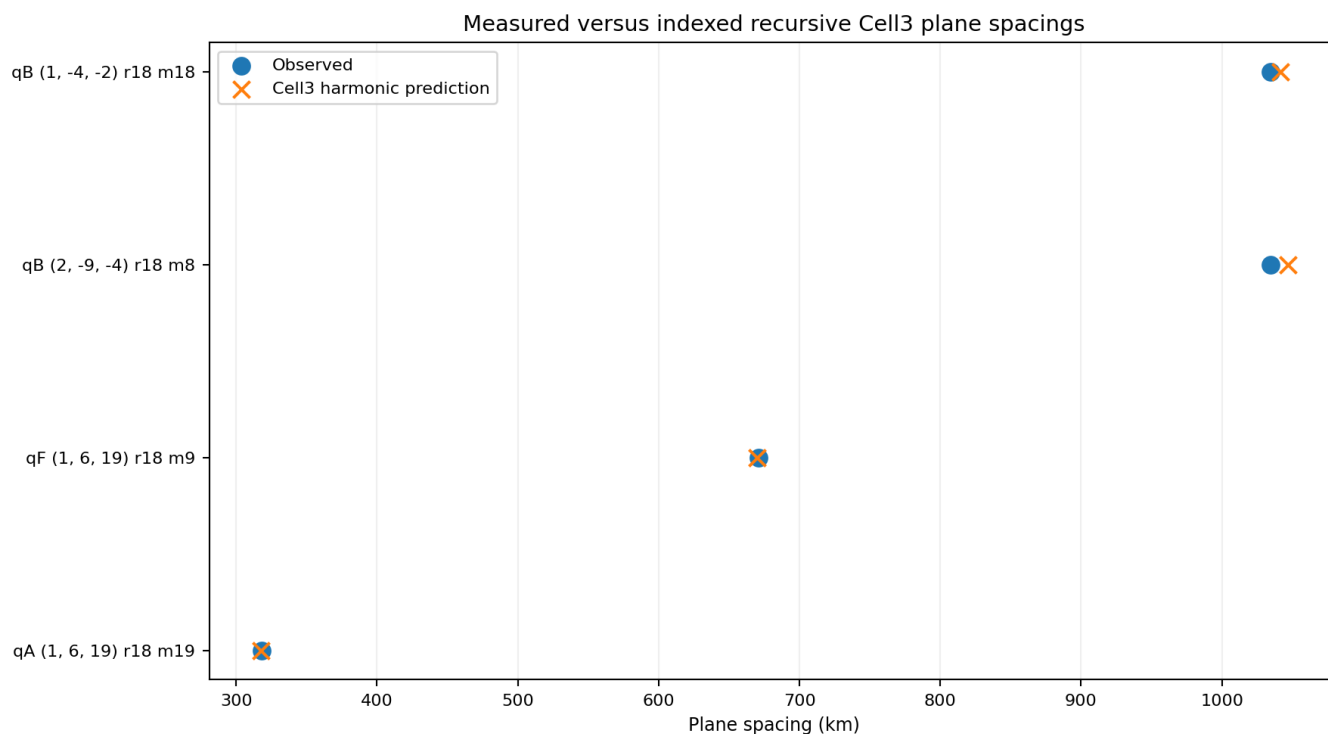


Figure 9. Observed spacings and recursive Cell3 harmonic predictions for qA, qF, and the two qB assignments. Each point uses the exact authoritative reciprocal basis, integer Miller packet, recursive level, and harmonic order.

Mode	hkl	r	m	Predicted spacing (km)	Observed spacing (km)	Spacing residual	Orientation residual	Zone value
qA	(1, 6, 19)	18	19	317.433284	317.864281	+0.135776%	1.782077 deg	0
qF	(1, 6, 19)	18	9	670.136932	670.885616	+0.111721%	4.182885 deg	0
qB	(2, -9, -4)	18	8	1046.724798	1034.800487	-1.139202%	1.360906 deg	0
qB	(1, -4, -2)	18	18	1041.406943	1034.800487	-0.634378%	2.746287 deg	1

8. Reciprocal-Vector Sum Closure and qC

The independently extracted broadband plane qC is close to the measured sum $qA + 2qB$. This is the macroscopic analog of the exact Cell3 reciprocal-vector addition law. It is evaluated both directly from the measured vectors and from integer Cell3 packets.

```
angle(qC, qA + 2 qB) = 0.736708467 deg
spacing residual = +0.202464774 percent
```

Reciprocal-vector closure: qC approximately $qA + 2 qB$

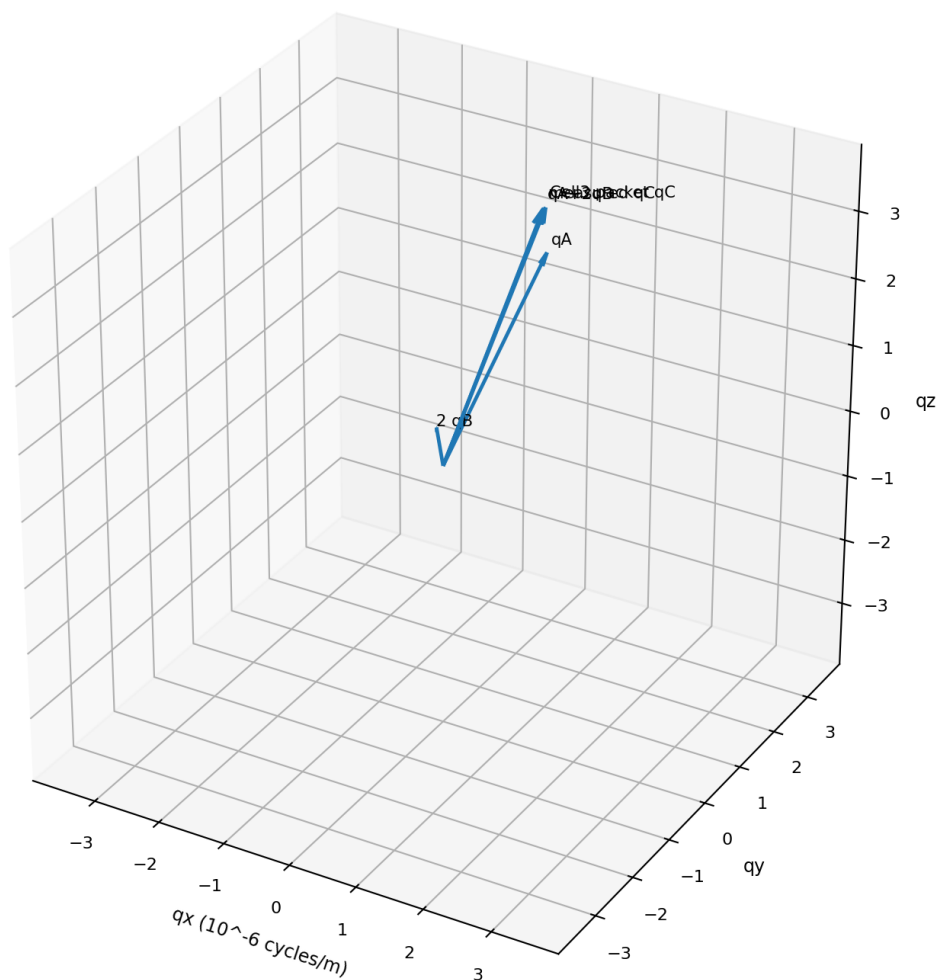


Figure 10. Three-dimensional reciprocal-vector closure. qA and 2qB add to a vector nearly coincident with measured qC. The exact Cell3 packet (17,-258,-433) at recursive level 18 is also plotted and nearly coincides with qC.

8.1 Integer packet derivation

```
qA packet: 19 (-1, -6, -19)
2 qB packet: 36 (1, -4, -2)
Sum: (17, -258, -433)
```

The resulting Cell3 packet predicts RA 47.862525 deg, Dec +68.908244 deg, and spacing 259.974612 km. The measured qC is RA 46.645302 deg, Dec +69.405455 deg, and spacing 260.276750 km. The orientation residual is 0.659366 deg and the spacing residual is +0.116218 percent.

This closure is stronger than a ratio comparison because it uses all three reciprocal-vector components. It connects the spectral-polarization mode, angular-variance mode, and broadband mode through one integer reciprocal packet.

9. Dimensionally Recovered Direct-Space Base Face

Two reciprocal vectors determine a minimum-norm direct-space dual. The measured vectors a_{exp} and b_{exp} are defined by $qA.a_{exp} = 1$, $qB.a_{exp} = 0$, $qA.b_{exp} = 0$, and $qB.b_{exp} = 1$. They are direct translations between neighboring intersections of the qA and qB plane families.

```
Q = [qA^T; qB^T]
D = Q^T (Q Q^T)^(-1)
a_exp = first column of D
b_exp = second column of D
```

Quantity	Measured exposed face	Harmonic Cell3 prediction	Residual
a length	319.393331 km	318.962296 km	+0.135137%
b length	1039.778275 km	1046.423190 km	-0.635012%
included angle	95.608657023 deg	95.612379801 deg	-0.003722778 deg
edge ratio b/a	3.255478981	3.280711244	-0.769110%
face area	3.305084e+11 m ²	3.321695e+11 m ²	-0.500099%

Dimensionally recovered exposed base face and harmonic Cell3 prediction

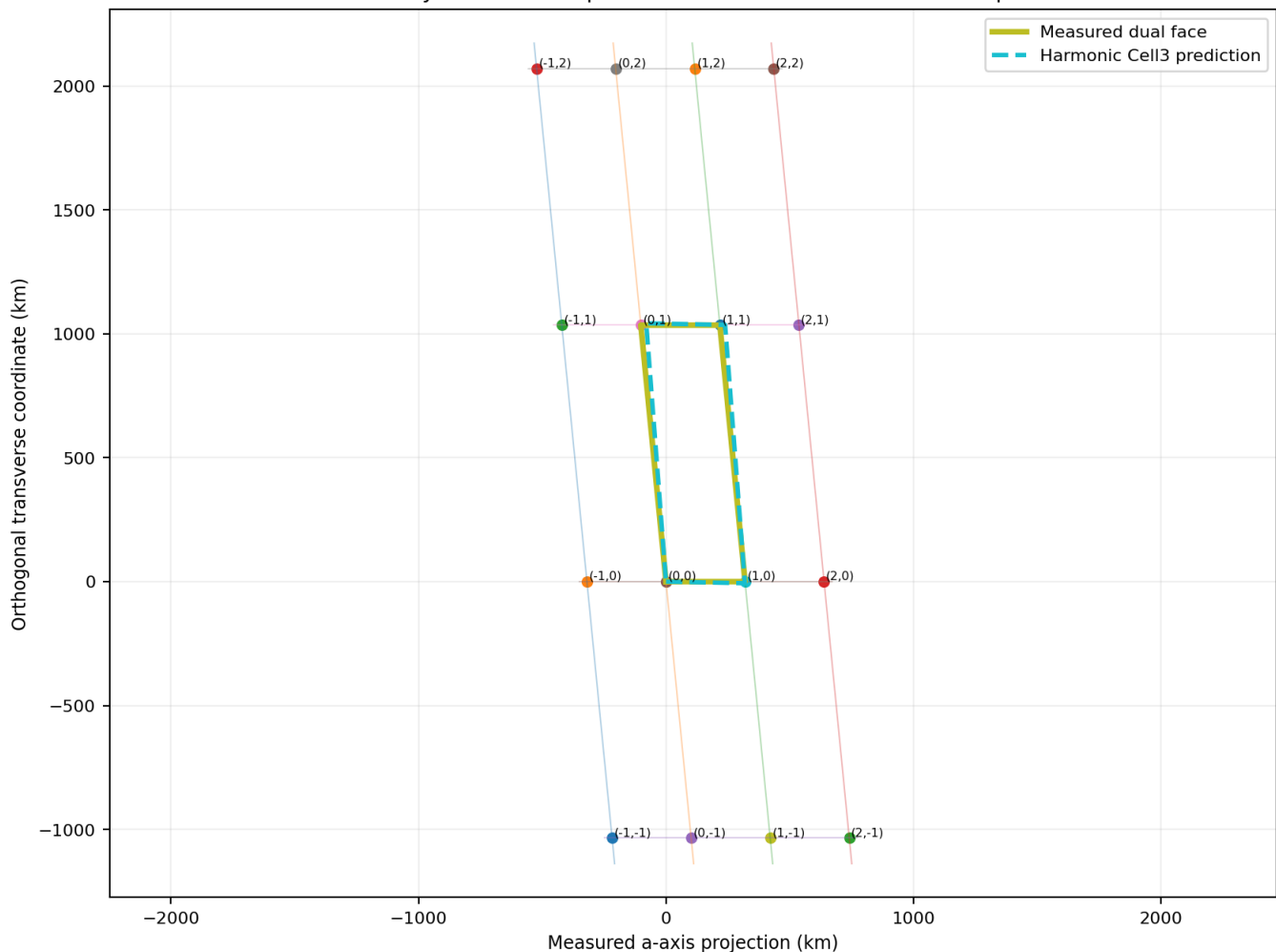


Figure 11. Dimensionally recovered exposed direct-space face with all 16 nodes in the local integer patch $-1 \leq i,j \leq 2$. The solid polygon is the measured base face. The dashed polygon is the harmonic Cell3 prediction generated from the qA and combination-optimized qB assignments. Axes are kilometres.

The measured included angle is 95.608657023 deg. Reversing either edge gives the equivalent acute angle 84.391342977 deg. The authoritative primitive Cell3 angle between a and b is 83.933483012 deg, a difference of +0.457859965 deg. This is a metric echo, not a direct identification of the exposed high-index face with the primitive ab face.

9.1 Plane and node interpretation

Every local node $N(i,j)$ is the intersection of one qA plane and one qB plane: $r(i,j) = i \text{ a_exp} + j \text{ b_exp}$. The qA phase at that node is exactly i cycles and the qB phase is exactly j cycles, apart from floating-point roundoff. The displayed local patch therefore has a direct, dimensionally exact relation to the measured reciprocal planes.

10. Spacing Ratios and Authoritative Cell3 Ratios

10.1 Primitive Cell3 area ratios

The authoritative primitive face areas and ratios are fixed by B. They are scale-invariant under uniform Cell3 recursion.

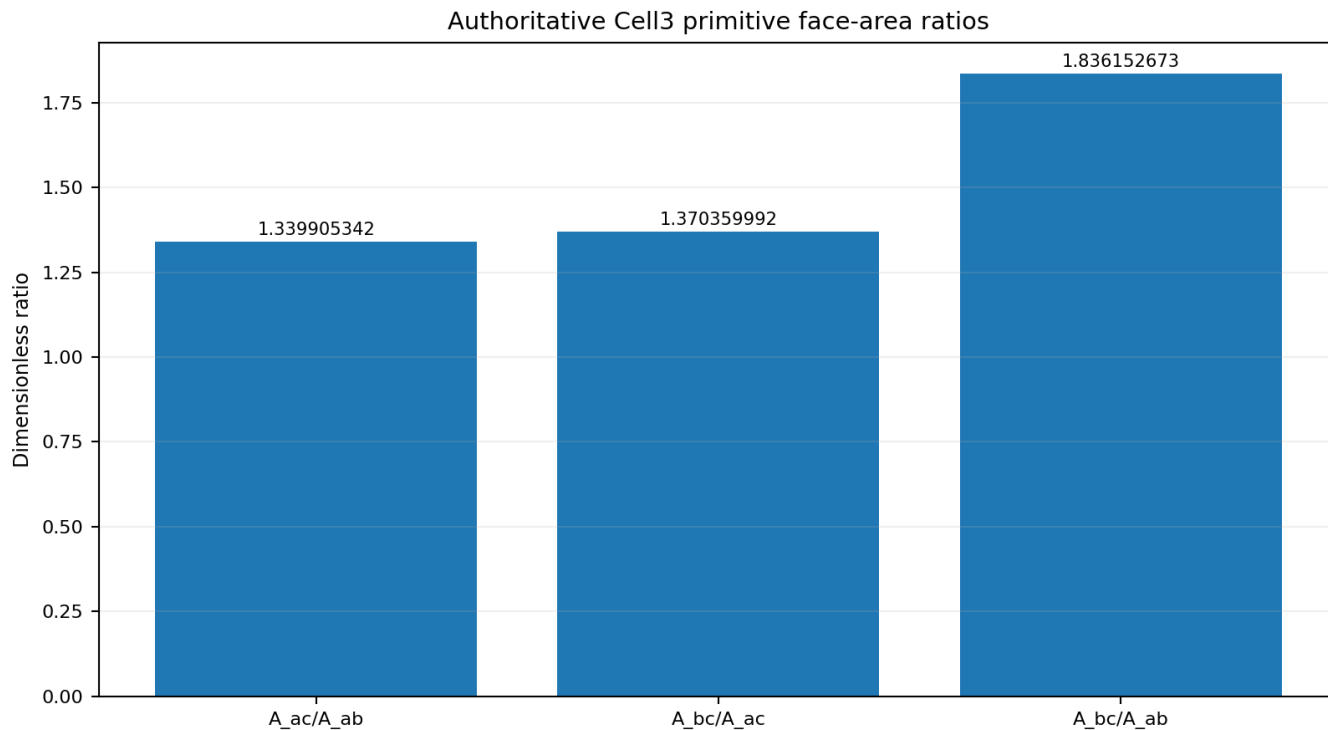


Figure 12. Authoritative Cell3 primitive face-area ratios. The exact multiplicative identity $(A_{ac}/A_{ab})(A_{bc}/A_{ac}) = A_{bc}/A_{ab}$ follows directly from cancellation of A_{ac} .

Ratio	Exact value
A_{ac} / A_{ab}	1.339905341991
A_{bc} / A_{ac}	1.370359991770
A_{bc} / A_{ab}	1.836152673424
Closure error	-2.220e-16

10.2 Direct edge ratios

Ratio	Value
Cell3 b/a	1.754777226618
Cell3 c/a	2.370939391865
Cell3 c/b	1.351134124549
Exposed b_exp/a_exp	3.255478980637

The exposed edge ratio does not equal a primitive Cell3 direct-edge ratio. The exposed face is the direct dual of high-index reciprocal planes with unequal harmonic multipliers. Its metric must therefore be computed from those indexed reciprocal vectors rather than compared as a simple scaled copy of a primitive face.

10.3 Harmonic and scaled-ratio relationships

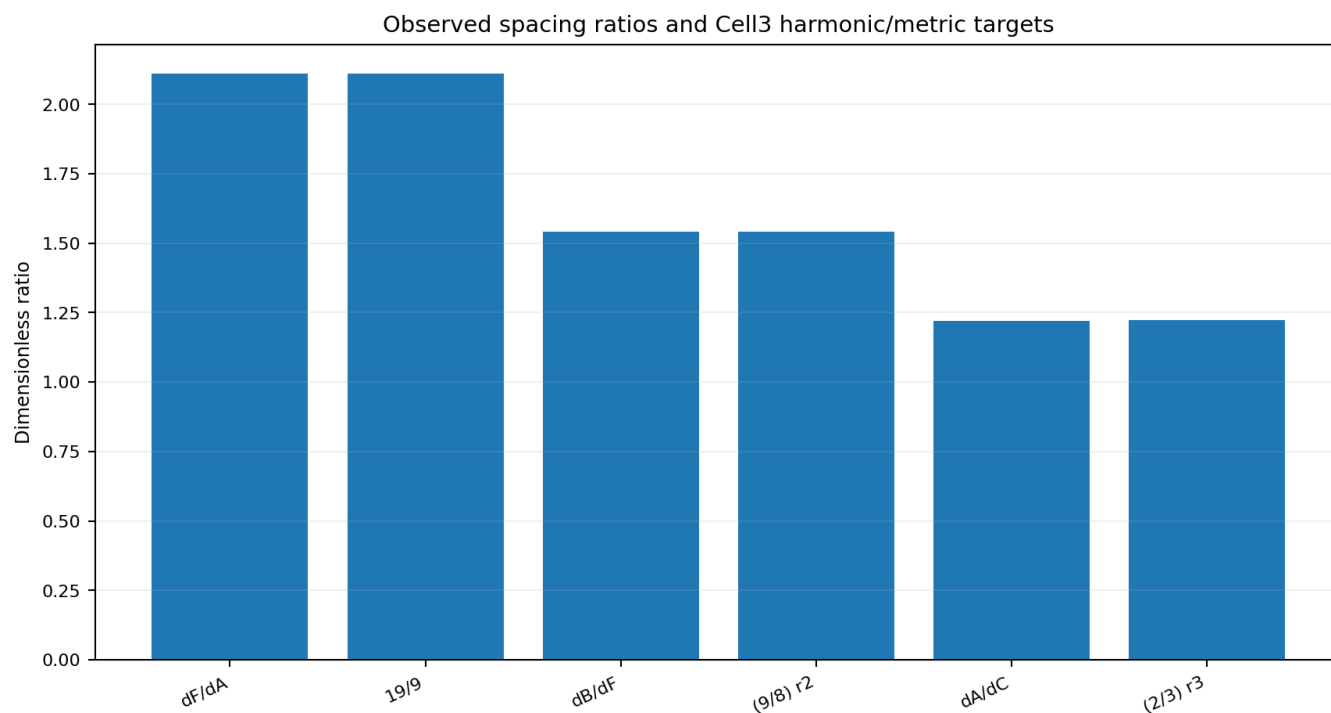


Figure 13. Observed spacing ratios compared with the primary 19/9 harmonic and two secondary Cell3 metric-scaled targets.

Relationship	Observed	Cell3 target	Residual
dF/dA	2.110603974823	2.111111111111	-0.024022%
dB/dF	1.542439519432	1.541654990741 = (9/8)(A _{bc} /A _{ac})	+0.050889%
dA/dC	1.221254995950	1.224101782282 = (2/3)(A _{bc} /A _{ab})	-0.232561%

The 19/9 relation is the primary harmonic because qA and qF share one indexed plane family. The two scaled area-ratio relations are secondary recurrences. Their integer prefactors were not independently fixed by one shared Miller family and should therefore be retained as structural clues rather than used as the primary indexing rule.

11. Geometric Synthesis

The complete comparison supports a layered interpretation. At the primitive level, Cell3 is a fixed oblique nanometre-scale basis with three unequal edges and seven primary reciprocal plane families. At the recursive level, direct lengths and plane spacings scale by 9^r . At the measured level, the spectrophotometer resolves high-index reciprocal directions and selected harmonic orders of those recursively enlarged plane families.

The qA and qF axes form one near-common family. Their 19:9 spacing ratio is reproduced to 0.02402 percent by the indexed family (1,6,19). qA and qB define a nearly orthogonal transverse pair whose intersection reconstructs the CMB-like longitudinal direction. In Cell3 direct coordinates, that line is close to [7,2,-1]. The exact zone law $7h + 2k - l = 0$ organizes the qA/qF family and the zone-preserving qB family.

The broadband qC mode is a reciprocal combination direction rather than an independent primitive axis. Its measured vector is close to qA + 2qB. The integer packet (17,-258,-433) reproduces its direction and spacing. The direct dual of qA and qB produces an exposed base face whose full metric is closely reproduced by the harmonic Cell3 construction.

Resolved structure. A measured two-dimensional high-index Cell3 reciprocal zone, two direct-space dual edges, a CMB-like zone direction, one common-family 19:9 harmonic pair, an integer qC sum packet, and a dimensionally recovered base face.

Unresolved structure. The direct period along the CMB-like zone direction. Without that longitudinal period, the exposed face cannot be closed into a finite three-dimensional macroscopic primitive cell, and the complete macroscopic face-area ratio set cannot be measured directly.

12. Educational Interpretation: How the Mapping Works

12.1 Why direct and reciprocal vectors differ

A direct vector joins equivalent lattice nodes. A reciprocal vector is perpendicular to a family of planes. In an oblique lattice, a reciprocal normal is generally not parallel to any direct edge. This is why comparing q_A directly with a , b , or c would be geometrically incorrect. The correct comparison uses $B^*(-T)$.

12.2 Why a high-index plane can have a simple zone

Large Miller indices do not imply an arbitrary direction. Planes with very different indices can all contain one simple direct-space line. The zone equation expresses this constraint. The measured q_A and q_B planes have high-index assignments, but their common line is close to the low-order direction $[7,2,-1]$.

12.3 Why recursive level and harmonic order are separate

Recursive level changes the physical size of the Cell3 copy. Harmonic order subdivides one indexed recursive plane interval. The same measured spacing can sometimes be represented by moving one recursive level and multiplying or dividing harmonic order by nine. This equivalence is exact because the hierarchy scale is nine.

12.4 Why vector closure is stronger than a scalar ratio

A spacing ratio compares only magnitudes. Reciprocal-vector closure compares magnitude and three-dimensional direction simultaneously. The q_C relation therefore carries more geometric information than a close ratio alone.

12.5 Why the longitudinal period remains difficult

The laboratory trajectory is dominated by an approximately constant CMB translation. The measured planes are nearly perpendicular to that translation. Tiny unresolved CMB-parallel components accumulate many cycles over the approximately 80-billion-metre trajectory. A third direct period requires an independent constraint along the zone axis rather than another transverse plane fit.

13. Reproduction Procedure

Step	Operation
1	Load the authoritative Cell3 basis B in ICRS nanometres and calculate $G = B^{-T}$.
2	Generate all 27 centered half-step nodes from $x = B(i,j,k)/2$ for i,j,k in $\{-1,0,+1\}$.
3	Load qA, qB, qC, and qF from the pic_id14E full-valid reciprocal-frame ledger.
4	Normalize every measured reciprocal vector and compute RA, Dec, spacing, and axial angle to CMB.
5	Calculate measured fractional Miller directions as $B^T n$ and normalize each by its largest absolute component.
6	Compare each measured direction to the seven primary Cell3 plane normals generated by $G(h,k,l)$.
7	Calculate $qA \times qB$, transform it through B^{-1} , and compare with the CMB direction in the same direct coordinates.
8	Fit a low-order direct zone direction and apply the exact zone equation $hu + kv + lw = 0$.
9	For each proposed high-index assignment, evaluate $q = (m/9^r)G(h,k,l) \times 10^9$ cycles/m and compare both direction and spacing.
10	Construct qC integer closure by adding the indexed qA and doubled qB packets.
11	Calculate the direct dual $D = Q^T(QQ^T)^{-1}$ for measured and predicted qA/qB pairs.
12	Generate the complete tables, node ledgers, dimensionally scaled figures, and machine-readable companion.

13.1 Dimensional checks

B is in nanometres. G is in inverse nanometres. The factor 10^9 converts inverse nanometres to inverse metres. Plane spacing is reported in metres or kilometres only after this conversion. Direct dual vectors obtained from reciprocal vectors in cycles per metre are automatically in metres. No arbitrary visual scale is used in the dimensional figures.

14. Conclusions

No.	Conclusion
1	The measured qA, qB, qC, and qF axes are not primitive low-index Cell3 plane normals.
2	The qA and qF modes are consistent with one high-index Cell3 family, (1,6,19), on recursive harmonic branches m=19 and m=9.
3	The measured dF/dA ratio differs from 19/9 by only -0.02402 percent.
4	qA and qB intersect along a line 0.036386 deg from the supplied CMB direction.
5	The intersection and CMB direction have nearly identical authoritative Cell3 direct-coordinate ratios.
6	The low-order direct zone [7,2,-1] gives the exact zone equation $7h + 2k - l = 0$.
7	The assignments (1,6,19) and (2,-9,-4) belong to that same exact zone.
8	The independently measured qC mode closes approximately as $qA + 2qB$.
9	The integer packet (17,-258,-433) predicts qC direction to 0.659366 deg and spacing to 0.116218 percent.
10	The complete measured direct-space face is reproduced by the harmonic Cell3 construction to sub-percent metric precision and 0.003723 deg in included angle.
11	The longitudinal period is not measured; therefore the result is an exposed two-dimensional Cell3 zone and base face, not a closed three-dimensional macroscopic primitive cell.

The comparison establishes a direct mathematical bridge from the authoritative nanometre Cell3 basis to the spectrophotometer-exposed large-scale plane structure. The bridge is carried by reciprocal orientation, integer zone geometry, the 9^{th} hierarchy, harmonic order, vector addition, and the recovered direct-space dual metric.

Appendix A. Complete Authoritative Cell3 Node Ledger

Address	Fractional coordinate	ICRS x (nm)	ICRS y (nm)	ICRS z (nm)	Radius (nm)
(-1,-1,-1)	(-0.5, -0.5, -0.5)	-0.176351350137	+0.118682429905	+0.637311498325	0.671826810838
(-1,-1,0)	(-0.5, -0.5, 0.0)	-0.181416008950	-0.076211421278	+0.292444294142	0.352482076442
(-1,-1,1)	(-0.5, -0.5, 0.5)	-0.186480667764	-0.271105272461	-0.052422910041	0.333198543969
(-1,0,-1)	(-0.5, 0.0, -0.5)	-0.158685962215	+0.222342355125	+0.363610955631	0.454785976632
(-1,0,0)	(-0.5, 0.0, 0.0)	-0.163750621028	+0.027448503942	+0.018743751447	0.167089839530
(-1,0,1)	(-0.5, 0.0, 0.5)	-0.168815279841	-0.167445347241	-0.326123452736	0.403600110809
(-1,1,-1)	(-0.5, 0.5, -0.5)	-0.141020574293	+0.326002280345	+0.089910412936	0.366398924014
(-1,1,0)	(-0.5, 0.5, 0.0)	-0.146085233106	+0.131108429162	-0.254956791247	0.321765879067
(-1,1,1)	(-0.5, 0.5, 0.5)	-0.151149891919	-0.063785422021	-0.599823995431	0.621855043707
(0,-1,-1)	(0.0, -0.5, -0.5)	-0.012600729108	+0.091233925963	+0.618567746878	0.625386652479
(0,-1,0)	(0.0, -0.5, 0.0)	-0.017665387922	-0.103659925220	+0.273700542695	0.293205445206
(0,-1,1)	(0.0, -0.5, 0.5)	-0.022730046735	-0.298553776403	-0.071166661489	0.307759169054
(0,0,-1)	(0.0, 0.0, -0.5)	+0.005064658813	+0.194893851183	+0.344867204183	0.396159882521
(0,0,0)	(0.0, 0.0, 0.0)	+0.000000000000	+0.000000000000	+0.000000000000	0.000000000000
(0,0,1)	(0.0, 0.0, 0.5)	-0.005064658813	-0.194893851183	-0.344867204183	0.396159882521
(0,1,-1)	(0.0, 0.5, -0.5)	+0.022730046735	+0.298553776403	+0.071166661489	0.307759169054
(0,1,0)	(0.0, 0.5, 0.0)	+0.017665387922	+0.103659925220	-0.273700542695	0.293205445206
(0,1,1)	(0.0, 0.5, 0.5)	+0.012600729108	-0.091233925963	-0.618567746878	0.625386652479
(1,-1,-1)	(0.5, -0.5, -0.5)	+0.151149891919	+0.063785422021	+0.599823995431	0.621855043707
(1,-1,0)	(0.5, -0.5, 0.0)	+0.146085233106	-0.131108429162	+0.254956791247	0.321765879067
(1,-1,1)	(0.5, -0.5, 0.5)	+0.141020574293	-0.326002280345	-0.089910412936	0.366398924014
(1,0,-1)	(0.5, 0.0, -0.5)	+0.168815279841	+0.167445347241	+0.326123452736	0.403600110809
(1,0,0)	(0.5, 0.0, 0.0)	+0.163750621028	-0.027448503942	-0.018743751447	0.167089839530
(1,0,1)	(0.5, 0.0, 0.5)	+0.158685962215	-0.222342355125	-0.363610955631	0.454785976632
(1,1,-1)	(0.5, 0.5, -0.5)	+0.186480667764	+0.271105272461	+0.052422910041	0.333198543969
(1,1,0)	(0.5, 0.5, 0.0)	+0.181416008950	+0.076211421278	-0.292444294142	0.352482076442
(1,1,1)	(0.5, 0.5, 0.5)	+0.176351350137	-0.118682429905	-0.637311498325	0.671826810838

Appendix B. Primary Plane Comparison Ledger

Plane	Spacing nm	RA	Dec	Mismatch qA	Mismatch qB	Mismatch qC	Mismatch qF
(100)	0.329546529696	355.201792	+1.869279	85.996170	66.906462	75.542581	84.916131
(010)	0.451597379721	84.132672	-29.408563	75.245658	29.694580	77.495965	77.844293
(001)	0.605097741518	258.000576	-21.000693	24.949144	74.439601	51.982262	27.363528
(110)	0.266175054578	27.349970	-15.252461	78.082454	78.743674	85.753979	78.715280
(101)	0.306442673082	326.873755	-8.694352	66.780932	59.961097	85.425297	68.446631
(011)	0.576944779286	106.780627	-75.703940	57.375300	31.348421	30.248009	54.706003
(111)	0.302545376075	3.082265	-28.568338	77.372278	84.966098	77.068079	77.195454

Appendix C. Exposed Plane and Assignment Ledger

Mode	q x	q y	q z	Spacing km	RA	Dec	Norm. Miller h,k,l
qA	3.287321093305e-07	2.168679499253e-06	2.255229490922e-06	317.864281	81.380625	+45.795586	(-0.039904420,-0.321646303,-1.000000000)
qB	3.066748724203e-07	-6.155152953415e-07	6.789419322445e-07	1034.800487	296.484362	+44.633516	(+0.222703711,-1.000000000,-0.473920797)
qC	9.277917496029e-07	9.826674478923e-07	3.596529571208e-06	260.276750	46.645302	+69.405455	(+0.040055461,-0.602920299,-1.000000000)
qF	1.779807172276e-07	9.748340072623e-07	1.113468312807e-06	670.885616	79.653162	+48.332024	(-0.032152027,-0.348869901,-1.000000000)

Mode	hkl	r	m	Pred. q x	Pred. q y	Pred. q z	Pred. spacing km	Orient. resid.	Spacing resid.
qA	(1, 6, 19)	18	19	-2.39132043e-07	-2.20436489e-06	-2.23780728e-06	317.433284	1.782077	+0.135776%
qF	(1, 6, 19)	18	9	-1.13273073e-07	-1.04417284e-06	-1.06001398e-06	670.136932	4.182885	+0.111721%
qB	(2, -9, -4)	18	8	2.95960138e-07	-6.25794407e-07	6.58409832e-07	1046.724798	1.360906	-1.139202%
qB	(1, -4, -2)	18	18	3.44779375e-07	-5.88956430e-07	6.75512584e-07	1041.406943	2.746287	-0.634378%

Appendix D. Exposed Direct-Space Node Ledger

Node	i	j	Measured x m	Measured y m	Measured z m	qA phase	qB phase	Predicted x m	Predicted y m	Predicted z m
(-1,-1)	-1	-1	-344469.340169	494637.081055	-868856.660641	-1.00000000 0	-1.00000000 0	-382802.474418	473372.935393	-872258.462102
(-1,0)	-1	0	-23580.891691	-241209.358896	-208024.244964	-1.00000000 0	-0.00000000 0	-13074.285037	-243490.051228	-205617.616359
(-1,1)	-1	1	297307.556787	-977055.798847	452808.170714	-1.00000000 0	1.0000000000	356653.904344	-960353.037850	461023.229383
(-1,2)	-1	2	618196.005265	-1712902.238798	1113640.586391	-1.00000000 0	2.0000000000	726382.093724	-1677216.024472	1127664.075126
(0,-1)	0	-1	-320888.448478	735846.439951	-660832.415677	0.0000000000	-1.00000000 0	-369728.189381	716862.986622	-666640.845742
(0,0)	0	0	0.000000	0.000000	0.000000	0.0000000000	0.0000000000	0.000000	0.000000	0.000000
(0,1)	0	1	320888.448478	-735846.439951	660832.415677	-0.00000000 0	1.0000000000	369728.189381	-716862.986622	666640.845742
(0,2)	0	2	641776.896956	-1471692.879902	1321664.831354	-0.00000000 0	2.0000000000	739456.378761	-1433725.973243	1333281.691485
(1,-1)	1	-1	-297307.556787	977055.798847	-452808.170714	1.0000000000	-1.00000000 0	-356653.904344	960353.037850	-461023.229383
(1,0)	1	0	23580.891691	241209.358896	208024.244964	1.0000000000	0.0000000000	13074.285037	243490.051228	205617.616359
(1,1)	1	1	344469.340169	-494637.081055	868856.660641	1.0000000000	1.0000000000	382802.474418	-473372.935393	872258.462102
(1,2)	1	2	665357.788646	-1230483.521005	1529689.076318	1.0000000000	2.0000000000	752530.663798	-1190235.922015	1538899.307844
(2,-1)	2	-1	-273726.665096	1218265.157743	-244783.925750	2.0000000000	-1.00000000 0	-343579.619306	1203843.089078	-255405.613024
(2,0)	2	0	47161.783382	482418.717792	416048.489927	2.0000000000	0.0000000000	26148.570074	486980.102457	411235.232718
(2,1)	2	1	368050.231859	-253427.722159	1076880.905604	2.0000000000	1.0000000000	395876.759455	-229882.884165	1077876.078461
(2,2)	2	2	688938.680337	-989274.162109	1737713.321281	2.0000000000	2.0000000000	765604.948836	-946745.870786	1744516.924203

Appendix E. Source, Figure, and Machine-Readable Manifest

Source file	SHA-256	Size bytes
K-Field_spectrophotometer_pic_id14E_analysis_exposed_plane_structure_1785932820.json	79e3aa2b187a9ad4fbd2b481de1b22ff7c53753c738678f3f593d3c0fc8e3c85	1545939
K-Field_ICRS_plane_scw_rajjectory_analysis_1785606737.json	03e6d901dbbe0801cada29a559d40b50c3cec5360f8e29f96d80e8e818cae3c3	47024
K-Field_student_introduction_1785346272.json	c16dc1d479efc074d0cd1f312193780a80e61e7e8ea7e014986d8258f51baa20	890582
pic_id14E_full24_hrs_20260805.csv	6f5e3c69c623fe02b854e121110f915861bab24e8b6f000b094e810f49b53f62	31949154
K-Field_spectrophotometer_pic_id14E_fullvalid_exploratory_analysis_1785932633.json	7ebd920092387642774c7135d618e3322791202cc0525b4f2cad3d3228ec4b6f	115320

Figure	SHA-256	Size bytes
authoritative_cell3_27_nodes.png	f921b42aec98ea102f85d67ef857a654b9affe5f0e17f4a8c3a143f60998729f	422878
authoritative_cell3_central_planes_all_nodes.png	e1b7d927e277461176ff8fdda36853f82cc474340626de85bc118d018388167d	66035
authoritative_cell3_face_area_ratios.png	6790c826be96bafcfb0aefacbd707b6db34643c8aef0cdb1011b4e460bfd483	50339
cell3_primary_plane_atlas_vs_measured.png	643d06958dbf447cc288c01b0f7ffcafe750ef2db9db5b7898ba39ae4d7d0f49	70670
common_cell3_zone_geometry_3d.png	67866c08cf1486ea50772510c4a5e6b446107a3adf562959f9aeecc1e27f9b70d	303552
exposed_base_face_measured_vs_cell3_prediction.png	91de1eb2e5d23b61e61d0760a6ca52afb13d748f50acb65f00a482a1d7018005	129996
indexed_plane_spacing_comparison.png	b0d3258dac0afa38fe6b4a2893916ca2ab0fa117a4871db7afe590b7521e8541	64954
low_index_plane_mismatch_ranked.png	dafde640d8b00fa500e75382d798eb19b7be5f9f2c31e9e7b652b4b5edfb35bb	66043
measured_normalized_miller_coordinates.png	6c62cf7078bd33262bb6474b30aa729f7e63da42892d77cba52fc9c366c3b5a6	51364
qA_qF_recursive_harmonic_stack.png	7e0fb8bbdc1e89865b874552d62b2b5d0af170ba269de11f499a75f50a116013	48629
qC_reciprocal_vector_closure.png	08143f27408da06704fccc5bf4509e696caa6329a8de390342916760702e600c	285045
spacing_ratio_comparisons.png	3d60020a67e93369d43383f1351429774efb64fb7af44ddd1cab92790940291	59435
title_cell3_exposed_plane_composite.png	6582f943e87e1a01ac3593a4845f233d038159e249a7d0f121445e0d03721ca1	513938

Appendix F. Archival Reproduction Record

Field	Value
Principal Investigator	P. Dwayne Esterline
Degrees	BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)
Organization	Krytronx, LLC.
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation epoch	1785937320
Generation UTC	2026-08-05T13:42:00Z
PDF	K-Field_cell3_to_spectrophotometer_planar_comparison_1785937320.pdf
JSON	K-Field_cell3_to_spectrophotometer_planar_comparison_1785937320.json
PDF engine	Python ReportLab
PDF fonts	Helvetica, Helvetica-Bold, Times-Roman, Courier only
Figure method	Python numerical computation and raster rendering from exact report ledgers
Authoritative Cell3 source	K-Field_ICRS_plane_scew_rajectory_analysis_1785606737.json
Spectrophotometer plane source	K-Field_spectrophotometer_pic_id14E_analysis_exposed_plane_structure_1785932820.json
Historical format references	K-Field Spectrophotometer Optimized Geometry; K-Field Spectrophotometer Hidden Structure Energy Analysis; K-Field pic_id14C Analysis
Machine-readable scope	Complete basis, 27 nodes, edges, plane atlas, measured planes, zone analysis, recursive assignments, qC packet closure, direct-space nodes, ratio comparison, source hashes, and figure manifest

END OF PROPRIETARY CONFIDENTIAL TRADE SECRET ARCHIVAL REFERENCE